

Promotion effects of Ce added Fe–Zr–K on CO₂ hydrogenation to light olefins

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Abstract Microspherical and uniform Fe–Zr–Ce–K catalysts were prepared by the microwave assisted precipitation followed by impregnation and promoted CO₂ hydrogenation toward advantageous synthesis of light olefins. The reducibility, surface basicity and surface atom composition of the catalysts was tuned as the Ce content was varied. The ceria spacer restrained the growth of the Fe₂O₃ crystallite. Ce addition weakened the interaction between Fe species and zirconia and the reduction of the Fe species became easier. On 35Fe–7Zr–1Ce–K at 320 °C and 2 MPa, the conversion of CO₂ and the selectivity of C₂–C₄ olefins were 57.34 and 55.67%, respectively. The ratio of olefin/paraffin was 7. The selectivity of CO and C₅⁺ hydrocarbons were 3 and 15.87%, respectively. The enhanced performance is attributed to intensified bifunctional activity toward CO hydrogenation as well as the reverse water–gas shift reaction.

Keywords Fe–Zr–K catalysts · Ce addition · Bifunction · CO₂ hydrogenation · Light olefins

Introduction

The recycling utilization of the green-house gas CO₂ to produce chemicals and fuels has been attracting considerable attention [1–7]. The hydrogenation of CO₂ to methanol, higher alcohols, formic acid, hydrocarbons and gasoline is a promising approach [5–9] considering the progress in obtaining non-fossil H₂. With ever-increasing demand on light olefins as the building blocks in polymer and fine

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chemicals industries, the synthesis of C_2 – C_4 olefins from CO_2 hydrogenation is of great interest. A two-step mechanism has been put forward for this reaction [10, 11]: first the reduction of CO_2 to CO via the reverse water–gas shift (RWGS), and then hydrogenation of CO to olefins via the Fischer–Tropsch synthesis (FTS).

Iron catalysts are active for the RWGS reaction and the FTS [2, 12, 13] and have been studied for CO_2 hydrogenation [2]. Potassium as a promoter improves the olefin selectivity and the promotion effects have been studied [14–16]. Nonetheless, there is high production of CH_4 , CO and heavier hydrocarbons from CO_2 hydrogenation over bulk iron catalyst. Fe–K/ Al_2O_3 –MgO [17], supported iron [18] and nanocomposite FeMnO [19] and ceria modified Fe/Mn/K [20] have been studied to control the product distribution. Fe–K/ ZrO_2 by co-impregnation shows enhanced selectivity of light olefins while heavier products and CO are high [18].

Metal/ ZrO_2 displays interesting catalytic behavior in CO_2 hydrogenation owing to interfacial effect between metal and ZrO_2 [21]. Our previous work has demonstrated that Fe–Zr–K by the microwave assisted precipitation functions high activity for CO_2 hydrogenation toward light olefins and low selectivity of CO. But C_5^+ products are higher, the chain growth probability remains larger [22].

To further optimize the product selectivity, it is necessary to tune the bifunctional activity of Fe–Zr–K toward subsequent hydrogenation of CO as well as the adsorption activation of CO_2 . Cerium-containing mordenites exhibit low temperature activity for the WGS reaction [23]. Cerium promoted Pt/ ZrO_2 is highly active toward the WGS at high temperatures, by stabilizing the microcrystal of the tetragonal phase and increasing the surface area [24]. In addition, CeO_2 shows the largest capacity of CO_2 adsorption in comparison with SiO_2 , Al_2O_3 and ZrO_2 [25]. In the presence of water vapor, CeO_2 even enables CO_2 to adsorb at low temperature, which is suitable to CO_2 hydrogenation with water as a product.

In the present study, the effects of cerium addition on the properties of Fe–Zr–K catalyst were studied in order to tune product distribution of CO_2 hydrogenation and to raise the yield of desired light olefins. The structure and catalytic performance of Fe–Zr–Ce–K were characterized and effect mechanism of Ce addition was discussed.

Experimental

Catalyst preparation

Ce added Fe–Zr samples were prepared by the microwave assisted homogeneous precipitation (MAHP) as described in literature [22]. The molar ratio of Fe/Zr/Ce included 50:10:1, 35:7:1, 15:3:1 and 5:1:1, where the ratio of Fe/Zr was 5:1 and the Ce content gradually increased. The molar ratio of (Fe+Zr+Ce)/urea was 1:6. The obtained samples were denoted in turn as 50Fe–10Zr–1Ce, 35Fe–7Zr–1Ce, 15Fe–3Zr–1Ce, 5Fe–1Zr–1Ce, 5Fe–1Zr and Fe_2O_3 . Fe–Zr–Ce–K samples were prepared by incipient wetness impregnation of K_2CO_3 aqueous solution with a Fe/K molar ratio of 10:1 followed by drying at 120 °C overnight.

Characterization

Phases of the samples were analyzed on a Rigaku D/MAX 2200PC X-ray diffractometer with Cu K α radiation (40 kV, 30 mA). The crystallite size was calculated according to the Scherrer equation.

N $_2$ physical adsorption–desorption isotherms were analyzed on a JW-BK132F instrument. Prior to measurement, samples were degassed at 300 °C for 2 h. The BET surface area was calculated from the adsorption branch and the pore parameter calculation based on the BJH method.

Scanning electron microscopy (SEM) images were obtained from a JEOL-JSM-7500F microscope using an accelerating voltage of 1 kV.

H $_2$ temperature-programmed reduction (H $_2$ -TPR) of the samples was carried out on a TP5000 chemisorption instrument. Sample of 70 mg was pretreated in He (25 mL min $^{-1}$) at 350 °C for 1 h. Then the sample was reduced by 5% H $_2$ as the temperature raised from room temperature to 800 °C at a rate of 10 °C min $^{-1}$. The consumption of H $_2$ was recorded on a gas chromatography with thermal conductivity detector.

For CO $_2$ temperature-programmed desorption (CO $_2$ -TPD), sample of 200 mg was pretreated in a flow of 30% H $_2$ (30 mL min $^{-1}$) at 350 °C for 3 h. After purging by He (25 mL min $^{-1}$) for 1 h, the sample was pulse-adsorbed by CO $_2$ at 50 °C until saturation, followed by purge with He for 1 h. The TPD profiles were recorded as the temperature raised from 50 to 800 °C at 10 °C min $^{-1}$.

X-ray photoelectron spectra (XPS) were measured on a thermo scientific ESCALAB 250 spectrometer with Al K α X-ray source. The base pressure of the chamber was 2×10^{-9} mbar. The binding energies (BEs) were calibrated relative to adventitious carbon using the C 1 s peak at 284.8 eV.

Activity test

Catalytic activity for CO $_2$ hydrogenation was conducted on a flow-type microreactor (350 mm length $\times$ 8 mm i.d. $\times$ 12 mm o.d). The catalyst sample of 2 mL (20–40 mesh) was first reduced in situ at 400 °C for 8 h by 30% H $_2$ (GHSV = 1000 h $^{-1}$ and P = 0.1 MPa). The reaction conditions were H $_2$ /CO $_2$ (molar ratio) = 3/1, GHSV = 1000 h $^{-1}$, T = 320 °C and P = 2 MPa. The sampling interval was about 12 h during time-on-stream (TOS) of 96 h. The effluent gas was analyzed online on a GC-9160, using a ϕ 3 mm $\times$ 2 m TDX-01 column to separate C $_1$ products and a 50 m $\times$ 0.53 mm $\times$ 40 μ m Al $_2$ O $_3$ capillary column to C $_1$ –C $_5$ hydrocarbons. Liquid products were analyzed off-line on another GC-9160, with a ϕ 3 mm $\times$ 2 m Porapak Q column to separate aqueous phase products and a 50 m $\times$ 0.32 mm $\times$ 1.0 μ m HJ-1 capillary column to oil phase products. CO $_2$ conversion and product selectivity were calculated as follows:

$$\text{CO}_2\text{conv.} = (F_{\text{in}} \times y_{\text{in}}^{\text{CO}_2} - F_{\text{out}} \times y_{\text{out}}^{\text{CO}_2}) / (F_{\text{in}} \times y_{\text{in}}^{\text{CO}_2}) \times 100\%$$

$$\text{CH}_4(\text{CO}) \text{ sel.} = (\text{F}_{\text{out}} \times y_{\text{out}}^i) / (\text{F}_{\text{in}} \times y_{\text{in}}^{\text{CO}_2} \times \text{Conv.}_{\text{CO}_2}) \times 100\%$$

here F_{in} is the molar flow rate of feed gas, mol h^{-1} ; F_{out} was the mole flow rate of effluent, mol h^{-1} ; y_{out}^i is the molar fraction of CH_4 (CO) in the effluent. Hydrocarbon selectivity included gaseous phase and liquid phase products.

Results and discussion

Phases structure of the samples

XRD patterns of the samples before K impregnation are shown in Fig. 1. As the Ce content was increased, the diffraction peak strength of $\alpha\text{-Fe}_2\text{O}_3$ phase [22, 26] was gradually attenuated. The diffraction strength of tetragonal ZrO_2 phase [27] was diminished, too. CeO_2 phase [28] was only observed in 5Fe–1Zr–1Ce. The crystallite size of $\alpha\text{-Fe}_2\text{O}_3$ phase corresponding to the (104) crystal face is shown in Table 1. It tended to decrease as the CeO_2 content increased. For 35Fe–7Zr–1Ce and 5Fe–Zr–1Ce, it was 21.2 nm and 15.8 nm, respectively. It is inferred that the growth of Fe_2O_3 crystallite is restrained due to the existence of CeO_2 spacer. Furthermore, with decrease in the crystallization degree, more defects can be generated [29], which should be beneficial to activation of CO_2 . As seen in Fig. 2, the diffraction peaks of $\alpha\text{-Fe}_2\text{O}_3$ phase disappeared in the used samples while Fe_3O_4 phase [19] was found, suggesting the phase transformation of iron species. In contrast, CeO_2 phase did not change obviously before and after reaction. No diffraction peaks for potassium oxide were observed.

Texture properties of the samples

Fig. 3 shows the SEM images of the samples before K impregnation. Four samples were shaped in microspheric and uniform nanoparticles, and the size was in the

Fig. 1 XRD patterns of the fresh samples, **a** Fe_2O_3 , **b** 5Fe–1Zr, **c** 50Fe–10Zr–1Ce, **d** 35Fe–7Zr–1Ce, **e** 15Fe–3Zr–1Ce, **f** 5Fe–1Zr–1Ce

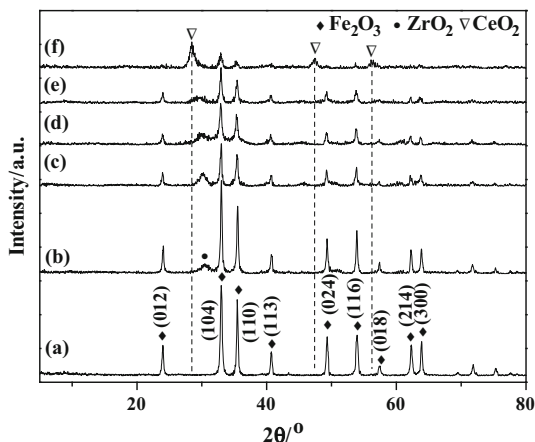
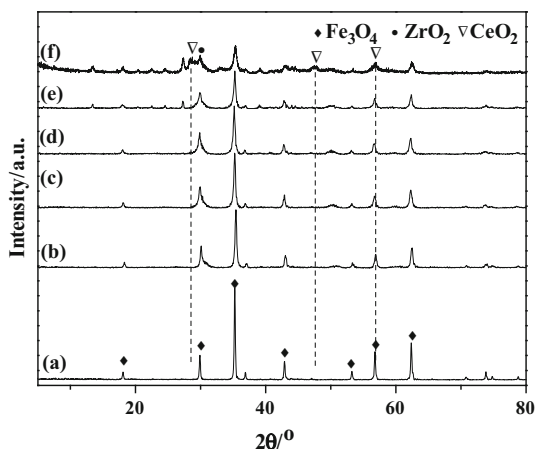


Table 1 Crystallite size of α - Fe_2O_3 phase in the samples*

Samples	Size (nm)
Fe_2O_3	24.9
5Fe–1Zr	28.1
50Fe–10Zr–1Ce	28.6
35Fe–7Zr–1Ce	21.2
15Fe–3Zr–1Ce	23.7
5Fe–1Zr–1Ce	15.8

* α - Fe_2O_3 crystallite size calculated by Scherrer equation from XRD results

Fig. 2 XRD patterns of the samples after reaction, **a** Fe–K, **b** 5Fe–1Zr–K, **c** 50Fe–10Zr–1Ce–K, **d** 35Fe–7Zr–1Ce–K, **e** 15Fe–3Zr–1Ce–K, **f** 5Fe–1Zr–1Ce–K

range of 25–60 nm. On the one hand, during the MAHP a consistent pH value is provided by the hydrolysis of urea. On the other hand, nucleation would be faster than growth of the crystal particle at hydrothermal temperature of 170–180 °C, enhancing the formation of regular particles [30]. As the CeO_2 content was increased, the particles tended to be smaller.

The samples had type IV isotherms of N_2 adsorption–desorption with the H_3 -type hysteresis loop, indicating the existence of mesopores [18]. The pore parameters are summarized in Table 2. Fe–Zr–Ce had larger surface area than Fe–Zr and Fe_2O_3 . As the CeO_2 content was increased, both the surface area and the total pore volume decreased. 35Fe–7Zr–1Ce possessed the largest surface area. The mesopore volume was much larger than the micropore volume. The mesopore diameter was about 2.4 nm.

Reduction behavior

The reduction of hematite can be essentially different depending on preparation conditions. A three-step mechanism $3\text{Fe}_2\text{O}_3 \rightarrow 2\text{Fe}_3\text{O}_4 \rightarrow 6\text{FeO} \rightarrow 6\text{Fe}$ and a two-step pathway $3\text{Fe}_2\text{O}_3 \rightarrow 2\text{Fe}_3\text{O}_4 \rightarrow 6\text{Fe}$ have been proposed [17, 31]. As seen in Fig. 4, the reduction of the samples fell into two stages. For Fe_2O_3 , the sharp reduction peak at 280–420 °C and the broad one at 450–800 °C were assigned to

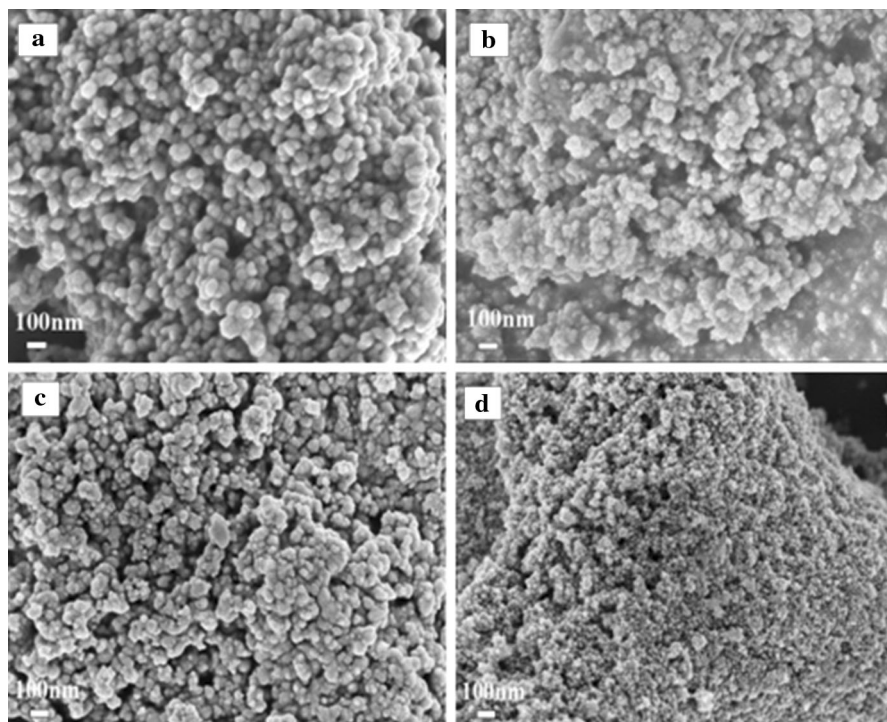


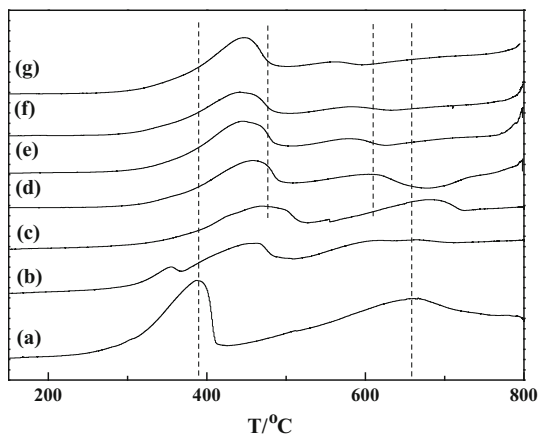
Fig. 3 SEM photographs of the samples, **a** Fe_2O_3 **b** 5Fe-1Zr **c** 35Fe-7Zr-1Ce **d** 15Fe-3Zr-1Ce

Table 2 Pore parameters of the samples

Samples	Surface area ($\text{m}^2 \text{g}^{-1}$)	Total pore volume ($\text{cm}^3 \text{g}^{-1}$)	Mesopore volume ($\text{cm}^3 \text{g}^{-1}$)	Micropore volume ($\text{cm}^3 \text{g}^{-1}$)	Mesopore diameter (nm)
Fe_2O_3	22	0.096	0.088	0.008	2.54
5Fe-1Zr	53	0.210	0.190	0.020	2.25
35Fe-7Zr-1Ce	78	0.191	0.163	0.028	2.43
15Fe-3Zr-1Ce	77	0.191	0.161	0.030	2.37
5Fe-1Zr-1Ce	72	0.156	0.128	0.028	2.39

initial reduction of $\alpha\text{-Fe}_2\text{O}_3\text{-Fe}_3\text{O}_4$ and then the reduction of Fe_3O_4 to $\alpha\text{-Fe}$, respectively [32, 33]. Two low-temperature peaks of Fe-K were ascribed to the reduction of $\alpha\text{-Fe}_2\text{O}_3$ to Fe_3O_4 : the reduction at around 350 °C of small crystallite particle of Fe_2O_3 and the reduction at around 460 °C of bulk Fe_2O_3 or iron species with strong interaction between Fe^{3+} and potassium [17, 33]. The third broad peak at 510–690 °C was assigned to the reduction of $\text{Fe}_3\text{O}_4\text{-}\alpha\text{-Fe}$. These three peaks were decreased after K impregnation because potassium suppresses the adsorption of H_2 [34] and then the reduction degree of Fe_2O_3 . Two reduction peaks of 5Fe-1Zr-K

Fig. 4 H_2 -TPR profiles of fresh samples, **a** Fe_2O_3 , **b** Fe–K, **c** 5Fe–1Zr–K, **d** 50Fe–10Zr–1Ce–K, **e** 35Fe–7Zr–1Ce–K, **f** 15Fe–3Zr–1Ce–K, **g** 5Fe–1Zr–1Ce–K

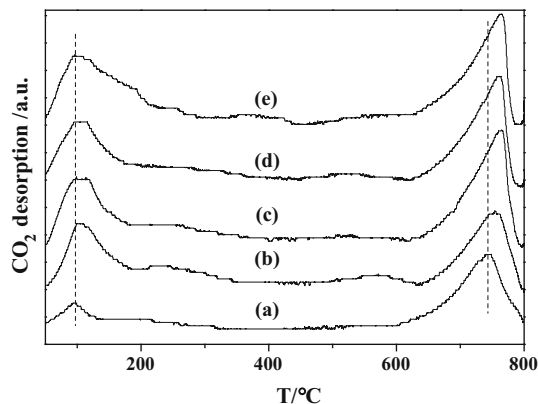


shifted to high temperatures, implying the inhibition effect on the reduction of Fe_2O_3 and the formation of iron phase through FeO phase. There exists strong interaction between iron species and zirconia [35, 36]. As the Ce content was increased, however, the two reduction peaks of Fe–Zr–Ce–K shifted to lower temperatures compared with Fe–Zr–K, indicating a small amount addition of Ce abates this interaction and facilitates the reduction of iron species.

CO_2 adsorption

As seen in Fig. 5, Fe–K sample appeared two CO_2 desorption peaks at 120–170 °C and 610–800 °C, respectively. For 5Fe–1Zr–K, the peaks slightly shifted to high temperatures and the peak area increased. In addition, two new small desorption peaks were produced at 200–320 °C and 510–620 °C, implying that the adsorption of CO_2 relating to the surface basicity [37], is intensified due to the addition of ZrO_2 [22]. After the addition of Ce, the peak temperature and areas of the high-temperature desorption peaks was obviously increased, suggesting that addition of

Fig. 5 CO_2 -TPD profiles of the samples, **a** Fe–K, **b** 5Fe–1Zr–K, **c** 35Fe–7Zr–1Ce–K, **d** 15Fe–3Zr–1Ce–K, **e** 5Fe–1Zr–1Ce–K



Ce enhanced the adsorption strength of CO_2 and the surface basicity [25]. 35Fe–7Zr–1Ce–K showed the strongest high-temperature desorption peak. A good correlation can be found between CO_2 and TPD results and the CO_2 hydrogenation activity (see Table 4).

Iron chemical state

Change in iron chemical state of the samples is shown in Fig. 6. Binding energy (BE) at around 710.9 eV and 724.5 eV was ascribed to Fe ($2p_{3/2}$) and Fe ($2p_{1/2}$), respectively, which is typical for Fe^{3+} ions in an environment of oxide state. The small satellite peak at about 718.9 eV characterizes the existence of Fe_2O_3 [19]. The shift in Fe(2p) BEs of 5Fe–1Zr compared with Fe_2O_3 illustrates the interaction between iron and zirconia as revealed by H_2 -TPR (see Fig. 4). As the Ce addition was increased, the BEs of Fe(2p) peak decreased, indicating strong electronic effect between iron species and CeO_2 and increase in electron density in the iron nucleus. The calculated surface element composition of the samples is presented in Table 3. It is found that the surface atom ratios of Fe/Zr and Fe/Ce were lower than the preparation ratios, implying that Zr and Ce are enriched on the sample surface.

Catalytic activity

Uniform microspheres of Fe–Zr–Ce–K was expected to intensify bifunctional activity toward the RWGS and the F–T reactions during CO_2 hydrogenation. The catalytic performance of the samples for CO_2 hydrogenation is shown in Table 4. Products included CO, hydrocarbons and low amount of oxygenates such as methanol and ethanol. Compared with 5Fe–1Zr–K, Fe–Zr–Ce–K samples obviously improved the activity. As the Ce content was increased, the conversion of CO_2 , the selectivity of $\text{C}_2^--\text{C}_4^--$ and the ratio of O/P first increased and then decreased. The selectivity of C_5^+ decreased monotonously. Over 35Fe–7Zr–1Ce–K, the conversion of CO_2 and the selectivity of $\text{C}_2^--\text{C}_4^--$ reached 57.34 and 55.62%, respectively. The O/P ratio in the $\text{C}_2\text{--C}_4$ hydrocarbon products was 7.08. The selectivity of CO

Fig. 6 Fe(2p) binding energy of the samples, **a** Fe_2O_3 , **b** 5Fe–1Zr, **c** 50Fe–10Zr–1Ce, **d** 35Fe–7Zr–1Ce, **e** 5Fe–1Zr–1Ce

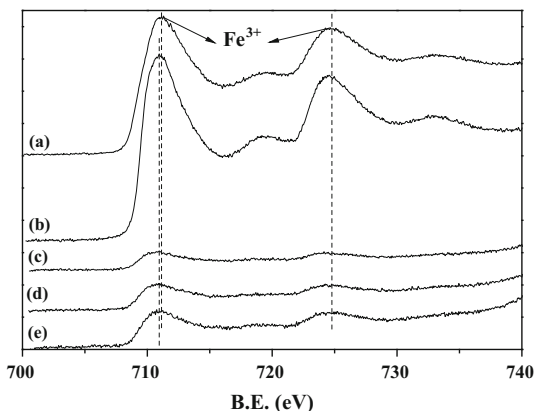


Table 3 Surface atom composition of the samples

Samples	Content (mol%)*						
	Fe	Zr	Ce	O	C	Fe/Zr	Fe/Ce
5Fe–1Zr	17.59	7.42	0	51.22	23.77	2.4	–
35Fe–7Zr–1Ce	11.38	3.65	1.03	46.62	37.32	3.1	11.1
15Fe–3Zr–1Ce	9.45	4.43	1.90	47.89	36.33	2.1	5.0
5Fe–1Zr–1Ce	15.74	3.64	7.14	53.54	19.94	4.3	2.2

*Calculated from the XPS data

Table 4 Catalytic activity of the samples for CO₂ hydrogenation

Catalysts	CO ₂ Conv./%	Sel./%			Hydrocarbon distribution ^c C/%				O/P ^d
		CO	HC ^a	Oxy ^b	CH ₄	C ₂ ⁼ –C ₄ ⁼	C ₂ ⁰ –C ₄ ⁰	C ₅ ⁺	
Fe–K	48.33	7.45	89.17	3.38	16.12	50.53	6.69	26.67	7.55
5Fe–1Zr–K	54.36	3.00	96.58	0.42	19.78	53.63	8.33	18.26	6.44
50Fe–10Zr–1Ce–K	55.31	3.25	96.10	0.65	20.78	53.67	8.08	17.46	6.64
35Fe–7Zr–1Ce–K	57.34	3.05	96.29	0.66	20.64	55.62	7.86	15.87	7.08
15Fe–3Zr–1Ce–K	48.98	3.43	94.73	1.84	21.43	51.76	11.12	15.69	4.66
5Fe–1Zr–1Ce–K	28.16	10.5	88.69	0.81	46.75	25.21	22.57	5.45	1.12

Reaction conditions: H₂/CO₂ = 3/1, GHSV = 1000 h^{–1}, 320 °C, 2 MPa, TOS = 84 h^aHC: hydrocarbons. ^bOxy.: oxygenates. ^cC₂⁼–C₄⁼: olefins; C₂⁰–C₄⁰: paraffins; C₅⁺: hydrocarbons. ^d(C₂⁼–C₄⁼)/(C₂⁰–C₄⁰)

remained 3%. On the one hand, the microspheres of the catalysts become smaller with the CeO₂ spacer. Restraining the growth of Fe₂O₃ crystallite results in defects favorable of the CO₂ adsorptive activation. Moreover, strong surface basicity of 35Fe–7Zr–1Ce–K also boosts the adsorptive activation of CO₂. On the other hand, CeO₂ can lower the work function of iron by donating electron, enhancing the dissociative adsorption of CO while reducing the H₂ adsorption ability and thus changing the product distribution [38].

Furthermore, CO₂ conversion over 35Fe–7Zr–1Ce–K showed an induction period of about 36 h and was then stabilized well with time-on-stream. At the initial stage of reaction, the conversion of CO₂ tended to increase as the active phase of iron carbide was gradually formed. As the reaction time extends, the composition and structure of the Fe species change [39] and undergo transformation from iron oxide to iron carbide (Fe₅C₂) with higher reactivity in FTS [40]. Proper interactions between iron species and zirconia can stabilize the active sites for both the RWGS and FTS reactions [22].

Conclusion

Uniform microspheres of Fe–Zr–Ce–K catalysts were obtained by the microwave assisted homogeneous precipitation. Appropriate addition of Ce in Fe–Zr–K diminishes of the catalyst particles and increases the surface basicity, promoting the dispersion of active Fe species and the uptake and activation of CO₂. CeO₂ as structural promoter restrains the growth of Fe₂O₃ crystallite and enables the reduction easier. With intensified activity for both the RWGS reaction and F–T reaction during CO₂ hydrogenation, conversion of CO₂ and selectivity of light olefin are improved while selectivity for heavier products and CO suppressed.

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